Somesh Jaiswal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.00
EWING CHRISTIAN PUBLIC SCHOOL	Class XII (CBSE)	2022	90.60%
EWING CHRISTIAN PUBLIC SCHOOL	Class X (CBSE)	2020	91.20%

WORK EXPERIENCE

Samsung | Software Engineering Intern June 2026 - July 2026

Project Title - **Agentic AI framework Development for Workflow Automation on Android**

- **Direct Execution Routing**: Architected custom tool-chaining pipelines and a deterministic fast-path handler in Kotlin, bypassing iterative **ReAct agent loop**s and reducing overall task **latency by 65%.**
- **End-to-End Agentic Booking**: Automated manual doctor consultation bookings into an agentic workflow on Samsung Health, cutting user interaction time by 80% while maintaining a **96%+ task completion rate.**
- **Cross-App UI Automation**: Utilized **MCP server**-exposed tools and Android Accessibility Services to orchestrate automated WebView navigation and form filling across partner platforms like **PharmEasy and Tata 1mg**.
- **Interactive Slot Disambiguation**: Designed a UI-driven disambiguation flow for selecting appointment slots, resolving scheduling ambiguity without re-triggering agent reasoning loops and cutting **overhead by 40%**.

SCHOLASTIC ACHIEVEMENTS

- Currently ranked **2nd in the Computational Engineering (COE) department**, with an **SGPA of 9.43** and a **CGPA of 9.00**.
- Solved **1500**+ DSA and competitive programming problems across **Leetcode** (Knight:1950+ rating) and **Codeforces**.
- **Core Team Lead, BlockSpace** -- Mentored **50+ students** in blockchain, cryptocurrency, and smart contract technologies.

PROJECTS

Flight Booking & Management System GitHub

Node.js , Zod , Sequelize ORM , Mongoose ODM , Node-cron , JSON Web Token , RabbitMQ , Redis

- Developed 5 decoupled microservices (Auth, Flight, Booking, Reminder, Payment) following **MVC architecture**, configuring an **API gateway** for reverse-proxy request routing and IP-based rate limiting.
- Engineered robust **authentication** from scratch using **JWT** for secure session management and role-based control.
- Managed multi-database workflows with a **SAGA orchestration pattern**, using custom **Sequelize** transaction hooks to automatically release reserved airplane seats via compensating rollbacks if payments failed.
- Merged **RabbitMQ** for asynchronous message streaming and cached high-frequency flight search queries in Redis.
- Scheduled time-staged alerts using **node-cron** and enforced strict API input validation via **Zod** schema.

TripBuddy (Multi-Agent Travel Planner) | Python , LangGraph, MCP, LangSmith, Render **Website Link**

- Orchestrated a supervisor agent with input **guardrails** to dynamically route flight, hotel, weather, and budget tasks.
- Connected **local**, **remote**, and **custom Model Context Protocol** servers to pull live travel data from **AviationStack** (flights) , OpenWeatherMap (weather) , and **Tavily** (hotel search) for grounded, accurate recommendations.
- Established HITL approval flows with PostgreSQL checkpointing, integrated **LangSmith** for distributed execution tracing and latency monitoring, and deployed the production service on **Render**.

OriginAI - Multi-Agent AI Platform GitHub

Node.js , Zod , Multer , LangGraph , Redis , Mongoose , Docker , Firebase , Razorpay , LLMs , RAG, Morgan

- Architected a platform using **5 Node.js microservices** (Gateway, Chat, Auth, Billing, Agent) built with the **MVC pattern** and shared services to decouple core logic and seamlessly manage chat state.
- Implemented secure SSO via **Firebase** Auth and engineered a centralized API Gateway with Redis-backed session

management, achieving **sub-millisecond authentication** latency.
- Formulated a dynamic state machine via LangGraph to route complex queries to specialized agents, coupling Qdrant vector retrieval with a multimodal **RAG pipeline** to parse PDFs and images alongside AWS S3 artifact storage.

Built a SLM from scratch (configuration similar to GPT-2 model) | Python , PyTorch , Tiktoken , Numpy **GitHub**

- Used **tiktoken** (BPE) for **sub-word tokenization** and built a custom PyTorch DataLoader for data batching.
- Programmed core **GPT-2 components** (Self-Attention, Causal Masking, Multi-Head Attention, Layer Normalization, and Feed-Forward Networks) from scratch in PyTorch, assembling modular **transformer** blocks with residual connections for **autoregressive** generation.
- Applied **GELU** activations to solve the dead neuron problem and used the GPT approximation for the normal CDF.
- Trained using cross-entropy loss and enabled controlled text generation using **temperature** scaling and t**op-K sampling**.

TECHNICAL SKILLS AND COURSES

- **Languages**: C , C++ , Solidity , SQL , JavaScript, Python
- **Frameworks/Libraries** : Express.js, Node.js, Tensorflow, PyTorch, LangChain, LangGraph, Sequelize, Mongoose
- **Courses** : Intro to Programming, Data Structures & Applications, Operating Systems, Fundamentals of Machine Learning and Deep Learning, Differential Equations , Numerical Methods and Computing.